Relation Algebra

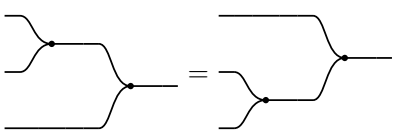
§2. Relations

(2a)

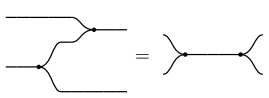
Rel is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartisian product, and converse $\circ \dashv \circ$, and

C: $(\triangleright : A \otimes A \to A, \multimap : \mathbb{I} \to A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \to A \otimes A, \multimap : A \to \mathbb{I})$

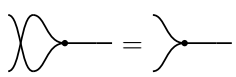
(2b)



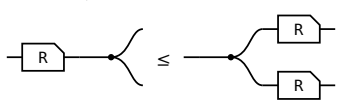
$\triangleright$ associative



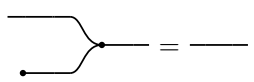
Frobenius, one half — the other is its $^\circ$



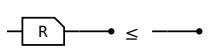
$\triangleright$ commutative



$R \triangleleft \leq \triangleleft (R \otimes R)$



$\multimap$ is its unit



$R \multimap \leq \multimap$

§2.1. ∀ object A, (A,◁,⊸)⊣(A,▷,⊶)

(2.1a)



▷ ◁ ≤ 𝟙 　　𝟙 ≤ ◁ ▷ 　　⊶ ⊸ ≤ 𝟙 　　𝟙 ≤ ⊸ ⊶

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In `Rel` it reads `{(a,a)}⊆A×A` — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.